

BHAVANA S

PROFESSIONAL SUMMARY

Highly skilled **Generative AI Data Scientist** with **6 years** of experience specializing in **LLMs, Python programming, and AI model fine-tuning**. Adept at **building, optimizing, and deploying large-scale AI models** to enhance accuracy and efficiency in AI-generated responses. Expertise in **hallucination reduction, bias detection, and prompt engineering** for **LLM-based applications**. Proficient in **Python, TensorFlow, PyTorch, and cloud platforms (AWS, GCP, Azure)** for scalable AI solutions. Passionate about driving **cutting-edge innovations in Generative AI**.

EDUCATION

MASTER'S IN DATA SCIENCE

University Of North Texas, Texas, United States

BACHELOR OF TECHNOLOGY COMPUTER SCIENCE AND ENGINEERING

GITAM Deemed To Be University, Andhra Pradesh, India

CERTIFICATIONS

AWS CERTIFIED DEVELOPER

Aws

DATA VISUALIZATION

Coursera

MACHINE LEARNING FOUNDATIONS

Coursera

PYTHON

Coursera

DATABASE MANAGEMENT ESSENTIALS USING SQL

Coursera

TECHNICAL SKILLS SUMMARY

- **Generative AI & LLM Fine-Tuning:** Expertise in optimizing and fine-tuning Large Language Models (GPT, BERT, BART, LLaMA) for various AI applications.
- **AI Chatbot Development:** Experience in building and deploying AI-powered chatbots using OpenAI GPT models, Microsoft Copilot, and transformer-based architectures.
- **Natural Language Processing (NLP):** Proficient in Named Entity Recognition (NER), text summarization (BART, GPT), TF-IDF vectorization, semantic analysis, and text classification.
- **AI & Machine Learning:** Skilled in supervised and unsupervised learning, including classification (Random Forest, Logistic Regression, SVM), clustering (K-Means, DBSCAN, OPTICS), and deep learning models (ANN, LSTM, Transformers).
- **Enterprise AI Integration:** Hands-on experience integrating LLM-based AI solutions into business workflows for automation, productivity enhancement, and intelligent decision-making.
- **Automation & AI Frameworks:** Strong expertise in AI-driven automation using Python-based frameworks, MLflow for model governance, and cloud-based AI services.
- **Cloud & AI Deployment:** Proficient in deploying AI models on Azure AI, AWS AI Services, and Google Cloud AI, with expertise in ETL pipelines, data warehousing (Snowflake, Databricks), and scalable AI solutions.
- **Programming & Development:** Advanced knowledge in Python, SQL, JavaScript, and C++, with experience in enterprise AI solutions, software development, and automation.
- **Visualization & Reporting:** Skilled in Power BI, Tableau, and Matplotlib for interactive dashboards and AI model performance monitoring.
- **AI Model Monitoring & Governance:** Expertise in bias detection, drift tracking, and explainability tools such as SHAP, LIME, and Fiddler AI for responsible AI implementation.

BUSINESS SKILLS

- Proven ability to extract meaningful business insights from complex datasets and drive data-driven decision-making.
- Strong analytical thinking and ability to translate business needs into technical solutions.
- Fast learner with a proactive approach to problem-solving and delivering results in a timely manner.
- Recognized for collaboration and leadership skills, with a track record of delivering high-impact solutions.

EXPERIENCE

GEN AI CONSULTANT

Tiger Analytics, Jersey City, NJ *(Jan 2024 – Present)*

- **Developed and fine-tuned LLMs** (GPT, BERT) for AI-driven automation, enhancing response accuracy by **30%**.
- Built **Python-based AI frameworks** for LLM fine-tuning, integrating Hugging Face models and OpenAI APIs.
- Implemented **automated bias detection pipelines** to ensure AI governance compliance.
- Optimized **prompt engineering techniques** for AI-generated responses, reducing hallucination rates.
- Designed **AI model monitoring dashboards** using **Power BI and Fiddler AI** for real-time drift detection.
- Developed **ML pipelines with Airflow and Kubernetes** for scalable model deployment and retraining.
- Led **LLM benchmarking and evaluation** to compare model performance across different architectures.

DATA SCIENTIST| AI MODEL TESTER

Telefónica, San Francisco, CA *(June 2023 – Dec 2023)*

- Deployed **GPT-based models** for text summarization and classification, improving response efficiency.
- Developed **real-time AI model monitoring dashboards** using **Azure ML and Fiddler AI**.
- Automated **bias and drift detection** techniques, ensuring AI compliance with governance frameworks.
- Fine-tuned **BERT models** for NLP-based ticket classification, improving accuracy by **25%**.
- Created **explainability pipelines using SHAP and LIME** to enhance model transparency and interpretability.
- Built **data ingestion and preprocessing pipelines** for AI model training using cloud-based ETL workflows.
- Conducted **A/B testing on AI-generated responses** to measure improvements in response accuracy and quality.

DATA SCIENTIST

TalentServe *(Mar 2021 – Dec 2022)*

- Developed an **ML model monitoring system** to track performance degradation and drift using **Fiddler AI**.
- Implemented **automated bias detection techniques**, improving model fairness by **20%**.
- Created a **CI/CD pipeline for AI model retraining and deployment** using **Airflow and Docker**.
- Designed an **NLP-based resume parsing system**, leveraging **BERT and TF-IDF** for skill extraction.
- Built an **explainable AI approach** to improve transparency in candidate ranking models.
- Automated **resume summarization and keyword extraction** using transformer-based models.
- Developed **predictive analytics models** to match candidates with job descriptions, improving hiring efficiency.
- Integrated **SQL-based data pipelines** for real-time data ingestion and processing.
- Deployed **machine learning models on cloud platforms** (AWS, GCP, Azure) for scalable processing.
- Designed **interactive Power BI dashboards** for AI model monitoring and performance visualization.
- Set up **ServiceNow-based alerts** to track anomalies and incidents in AI-driven hiring processes.
- Enhanced governance by implementing **AI compliance standards** in model validation workflows.
- Conducted **A/B testing experiments** to fine-tune NLP models for better resume classification.
- Established **pre-deployment AI model checks** to ensure compliance with risk mitigation policies.